CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 33 (2009) 511–516

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic and ab initio investigation of the Cu–Dy system

M. Palumbo [a,b,*], L. Battezzati [a], A. Pasturel [c,d], S. Gottlieb-Schönmeyer [e], W. Assmus [e]

[a] Dipartimento di Chimica I.F.M., Centro di Eccellenza NIS, Università di Torino, via Giuria 7/9, 10125 Torino, Italy

[b] National Institute for Materials Science, 1-2-1 Sengen, Tsukuba, 305-0047, Japan

[c] Sciences et Ingénierie des Matériaux et Procédés, INP Grenoble, 1130, rue de la Piscine, BP 75, 38402 Saint-Martin d'Hères Cedex, France

[d] Laboratoire de Physique et Modélisation des Milieux Condensés, Maison des Magistères, BP 166 CNRS, 38042 Grenoble Cedex 09, France

[e] Physikalisches Institut, Johann Wolfgang Goethe-Universität, Max-von-Laue-Strasse 1, 60438 Frankfurt am Main, Germany

ARTICLE INFO

Article history:
Received 3 October 2008
Received in revised form
3 April 2009
Accepted 8 April 2009
Available online 3 May 2009

Keywords:
Cu–Dy alloys
Ab initio calculations
Phase diagram
Thermodynamic modelling

ABSTRACT

A combined ab initio and thermodynamic study of the Cu–Dy system has been performed and a self-consistent thermodynamic database has been obtained. Density functional theory has been applied by using the VASP code in order to obtain the enthalpy of formation at 0 K of intermetallic compounds. Experimental information on the Cu–Dy phase diagram and thermodynamic properties of its alloys have been collected. Using these data, optimized parameters have been obtained by applying the CALPHAD approach. The present results reproduce the experimental data available reasonably well, although some features of the Cu–Dy system need to be further clarified.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

The phase diagrams of some Cu–RE (rare earth) systems have not been assessed yet and they deserve further investigation. The addition of RE elements could improve some properties of Cu based alloys, i.e. strength, electronic conductivity and magnetic properties [1,2]. Cu–RE systems are also of interest from a fundamental point of view because of the existence of complex metallic alloys. For RE = Dy and Yb, Cu_xRE phases with more than 7000 atoms in a monoclinic unit cell have been reported [3–5]. These are the most complex phases known to date. Such a complexity has recently generated new interest because of the possible implications on physical and chemical properties of these compounds [6]. A phase diagram assessment of these systems is thus desirable.

In the present paper we consider the Cu–Dy system by coupling thermodynamic modelling and ab initio density functional theory (DFT). This system has not been thermodynamically assessed to date. Experimental data on Cu–Dy alloys are limited and thus the contribution of ab initio calculations is helpful in clarifying some uncertainties on phase stability and estimating the enthalpy of formation of solid phases [7].

2. Literature information

A complete phase diagram investigation has been carried out by Franceschi [8]. A review of experimental data has been reported by Subramanian et al. [9]. The equilibrium phases in this system are: liquid, three terminal solid solutions (fcc Cu, bcc β-Dy and hcp α-Dy) and six intermetallic phases (CuDy, Cu_2Dy, and Cu_9Dy_2 which melt congruently; Cu_7Dy_2, Cu_5Dy and Cu_7Dy which melt peritectically). The solubility of Dy in fcc Cu and of Cu in both bcc and hcp Dy has been found to be negligible by Franceschi [8]. The homogeneity range of intermetallic compounds is also very limited.

Cheng et al. also studied the phase diagram [10]. In contrast to Franceschi, Cheng et al. reported the congruent formation of Cu_5Dy and the peritectic formation of both Cu_2Dy and CuDy. In addition, Cheng et al. did not observe the phases Cu_9Dy_2 and Cu_7Dy_2. However, the phase diagram reported by Franceschi shows the same pattern as for other RE–Cu systems, such as Cu–Gd [11], Cu–Er [12] and Cu–Y [13]. The Dy-rich region of the phase diagram has also been investigated by Copeland et al. [14]. They found a lower value (1048 K) for the eutectic temperature between liquid, CuDy and hcp Dy with respect to Franceschi (1063 K). Moreover, they found a high solubility of Cu in both bcc and hcp Dy, in contrast to Franceschi. Despite some Cu–RE phase diagrams showing a similar solubility (i.e. RE = Gd), most rare earth elements of the same series show negligible solubility (RE = La, Ce, Pr, Nd, Sm, Eu, Tb, Ho, Er, Tm, Yb). For this reason, we have adopted

* Corresponding author at: Dipartimento di Chimica I.F.M., Centro di Eccellenza NIS, Università di Torino, via Giuria 7/9, 10125 Torino, Italy. Tel.: +39 0 116707567; fax: +39 0 116707855.

E-mail address: mauro.palumbo@unito.it (M. Palumbo).

0364-5916/$ – see front matter © 2009 Elsevier Ltd. All rights reserved.
doi:10.1016/j.calphad.2009.04.001

Table 1
Crystalline structures, lattice parameters and atomic positions of Cu–Dy intermetallic compounds. Results from first principles calculations are also reported.

Phase name	Pearson symbol and prototype	Space group	Lattice parameters (nm)			Δ lattice parameters (exp. − calc.) (%)			Enthalpy of formation (kJ/mol of atoms)	Ref.
			a	*b*	*c*	*a*	*b*	*c*		
Cu_7Dy	hP8 (Cu_7Tb)	P6/mmm (no. 191)	0.4932		0.4156	0.2		0.2		[15]
			0.4922		0.4148				−13.3	Ab initio; this work
α-Cu_5Dy	cF24 ($AuBe_5$)	F43m (no. 216)	0.7027			0.3				[16]
			0.7022			0.2				[8]
									−15.7 ± 2.2	[21]
			0.70058						−19.4	Ab initio; this work
β- Cu_5Dy	hP6 ($CaCu_5$)	P6/mmm (no.191)	0.502		0.408	0.7		−0.3		[11]
			0.49839		0.40939				−19.0	Ab initio; this work
Cu_4Dy		P1	2.837	2.836	2.5658					[3]
Cu_9Dy_2	mC7448	C2	4.955	4.956	4.633				−12.7 ± 1.1	[3,8]
Cu_7Dy_2	mC7448	P1	2.133	2.129	1.891					[3]
Cu_2Dy	oI12 ($CeCu_2$)	Imma (no.74)	0.43	0.6792	0.73	1.2	0.4	0.3		[32]
			0.4303	0.6802	0.7289	1.3	0.5	0.1		[8]
									−16.4 ± 1.3	[20]
									−24.4 ± 2.3	[21]
			0.42488	0.6767	0.7281				−27.9	Ab initio; this work
CuDy	cP2 (CsCl)	Pm3m (no.221)	0.3461			−0.5				[33]
			0.346			−0.5				[34]
			0.3444			−1.0				[35]
			0.3461			−0.5				[36]
			0.3455			−0.7				[37]
			0.3455			−0.7				[8]
									−12.5 ± 2.0	[20]
			0.34789						−26.1	Ab initio; this work
CuDy	oP8 (BFe)	Pnma (no.62)	0.7087	0.4449	0.5346				−27.1	Ab initio; this work

the experimental results by Franceschi and neglected those by Copeland.

Table 1 summarizes the structural data on intermetallic compounds in the Cu–Dy system. Cu_7Dy was reported as stable at high temperature by [15]. Later, Franceschi determined its temperature stability range. Cu_5Dy was reported to exist in two crystal structures [16], a low-temperature cubic $AuBe_5$-type structure and a high-temperature hexagonal $CaCu_5$-type structure. The crystal structure of Cu_7Dy_2 was unknown for a long time, and Cu_9Dy_2 was assigned to have a tetragonal structure by Franceschi. Recently, Cerny et al. [3] have reported a new compound, Cu_4Dy, and structural models for Cu_xDy ($x = 4.5, 4.0, 3.5$) compounds. Interestingly, these structure models contain 7448, 1248 and 480 atoms in the monoclinic unit cell. The crystal structures of CuDy and Cu_2Dy have been fully determined and are reported in Table 1. A $Cu_{23}Dy_6$ phase stable at high pressure has been reported by Tsvyashchenko [17].

Contrary to what has been reported by Subramanian et al. [9], some thermodynamic data exist on the Cu–Dy system. The enthalpy of mixing of the liquid phase has been determined by Usenko et al. [18] at 1690 K. The enthalpy of mixing of the liquid phase has also been reported by Sudavtsova et al. [19]. Unfortunately, the original experimental data are not shown in the paper and these authors report the coefficients of a polynomial fitting of the original data without clearly reporting the polynomial form. Thus, these data have not been used in the present thermodynamic assessment. The enthalpy of formation of Cu_9Dy_2, Cu_2Dy and CuDy phases has been measured by Sommer et al. and by Meschel et al. [20,21]. Estimated values have also been determined using the Miedema semiempirical approach by Subramanian et al. [9] and ab initio calculations (see below). All these data on the enthalpy of formation have been used in the assessment.

3. Modelling

3.1. First principles calculations

First principles calculations have been carried out in order to determine the thermodynamic properties of intermetallic compounds in the Cu–Dy system. A complete list of these phases is reported in Table 1. Calculations have been performed on most Cu–Dy intermetallic compounds and on pure components, except where the structure was not fully resolved.

A full structure minimization at the ground state (0 K) has been carried out using Density Functional Theory (DFT) as implemented in the Vienna Ab initio Simulation Package (VASP) with plane wave basis sets [22–24]. The calculations employed the generalized gradient approximation (GGA) of Perdew and Wang [25] and valence electrons were explicitly represented with projector augmented wave (PAW) pseudopotentials. The plane wave cut-off energy was set to 273 eV for all Cu–Dy compounds. Odd-sized *k*-point meshes were created by a Monkhorst–Pack scheme [26]. The ground state (0 K) structure was determined by minimizing the Hellmann–Feynman forces with the conjugate gradient algorithm, until all ionic forces were less than 0.01 eV/A.

For one compound, where there was only a partial occupancy of some atomic sites by the components, a proper supercell has been built. This was the case for Cu_7Dy with Cu_7Tb structure.

The energy of formation of each Cu–RE compound has been obtained from the following equation:

$$\Delta E^{Cu_xRE_y} = E^{Cu_xRE_y} - xE^{Cu} - yE^{RE} \quad (1)$$

where $E^{Cu_xRE_y}$, E^{Cu} and E^{RE} are the total energy for Cu_xRE_y, pure fcc Cu and pure hcp RE = Dy. At 0 K this corresponds also to the enthalpy of formation. Harmonic and anharmonic contributions at higher temperature have been neglected in the present calculations.

3.2. Thermodynamic modelling

In the framework of the CALPHAD methodology, the description of the Gibbs free energy of a phase φ is based on the following expression:

$$G^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{ex}G^{\varphi} \quad (2)$$

where the first term (${}^{ref}G^{\varphi}$) is the contribution of the pure components, the second one (${}^{id}G^{\varphi}$) is the ideal mixing contribution and the third one is the contribution due to non-ideal interactions between the components, also known as the excess Gibbs energy of mixing. Each of these terms could depend on temperature, pressure and composition. Empirical parameters in them depend on the model adopted.

In the following, the pure solid elements in their stable phases at 298.15 K were chosen as the reference state for the system (SER, Standard Element Reference), except where otherwise specified.

According to the CALPHAD approach, the model parameters need to be determined by a computer-assisted statistical procedure to obtain the best fit of available experimental data. This fitting procedure is based on numerical techniques [27]. In the assessment procedure, the differences between experimental and computed values are minimized by a least square method. The optimised parameters are provided in the Appendix. The Thermo-Calc software and its Parrot module were used for thermodynamic optimization. The enthalpy of formation at 0 K from ab initio calculations was also used in the optimization with the available experimental data.

3.2.1. Pure elements

The thermodynamic functions of pure elements (lattice stabilities) have nowadays been well assessed and they are collected in a database compiled by Dinsdale [28]. These functions were used in the present assessment.

3.2.2. Liquid and solution phases

The liquid phase has been described by applying Redlich–Kister polynomials [29]. Accordingly, the Gibbs free energy of the liquid phase is described as

$$\begin{aligned} {}^{ref}G^{liq} &= x_{Cu}G^{liq}_{Cu} + x_{RE}G^{liq}_{RE} \\ {}^{id}G^{liq} &= RT\left[x_{Cu}\ln(x_{Cu}) + x_{RE}\ln(x_{RE})\right] \\ {}^{ex}G^{liq} &= x_{Cu}x_{RE}\sum_{\nu} {}^{\nu}L^{liq}_{Cu,RE}(x_{Cu} - x_{RE})^{\nu} \end{aligned} \quad (3)$$

where ν is the order of the term considered and ${}^{\nu}L^{liq}_{Cu,RE}$ are interaction parameters which have been taken as temperature dependent (see Appendix).

According to experimental determination, the terminal solid solubility in the hcp α-Dy and fcc Cu phases has been neglected. For the bcc β-Dy, an ideal solution model has been considered since a solubility range was suggested by [14].

3.2.3. Intermetallic compounds

The intermetallic Cu_xRE_y compounds present no solubility range according to the available experimental data. Thus, they have been modelled as stoichiometric compounds:

$$G^{Cu_xRE_y} = \frac{x}{x+y}G^{SER}_{Cu} + \frac{y}{x+y}G^{SER}_{RE} + a + bT \quad (4)$$

where G^{SER}_{Cu} and G^{SER}_{RE} are the Gibbs free energies of pure Cu and RE (=Dy) elements respectively in their stable state at SER (Standard Element Reference, 298.15 K and 1 bar) and a and b are optimised parameters.

Due to the complete lack of thermodynamic experimental data, the Cu_4Dy phase reported by Cerny et al. [3] was not included in the present assessment.

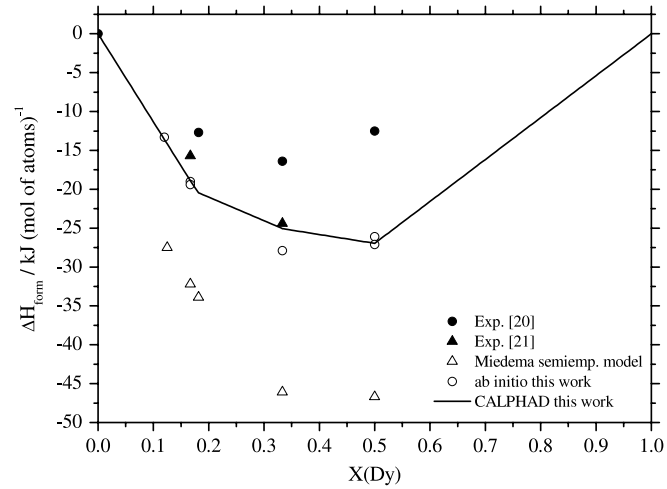

Fig. 1. Calculated enthalpy of formation of Cu–Dy intermetallic compounds according to the present thermodynamic assessment (full line), ab initio DFT calculations (open circles) and Miedema semi-empirical model (open triangles). Experimental data from [20,21] are also reported (full circles and full triangles respectively). Values are referred to SER reference states at 298.15 K and 1 bar.

4. Results and discussion

Results from ab initio calculations of lattice parameters and enthalpies of formation for Cu–Dy compounds are reported in Table 1. The calculated lattice parameters for relaxed structures are in good agreement with experimental data, with a maximum difference of about 1%. The calculated enthalpies of formation at 0 K are also reported in Fig. 1. They show the expected convex hull for compounds which are stable at low temperature (Cu_5Dy with $CaCu_5$ structure, Cu_2Dy and CuDy with CsCl structure), in agreement with their thermodynamic stability. Furthermore, the enthalpy of formation point for Cu_7Dy in Fig. 1 lies above the dotted line joining the enthalpy of formation point for Cu_5Dy and the one for pure fcc Cu, which is in agreement with experimental findings that report Cu_7Dy as stable only at high temperatures. The enthalpy of formation of Cu_5Dy with $AuBe_5$ structure is very close to the value for $CaCu_5$ structure, but lower and in agreement with the experimental higher stability of the former phase. Calculated values for the enthalpy of formation of CuDy with CsCl and BFe structures are also very similar, but in this case the BFe structure is slightly more stable than the CsCl structure at 0 K. In contrast, experimental findings report CuDy with the CsCl structure as stable. However, since the difference in both cases is limited (1 kJ/mol of atoms) and experiments refer to room temperature instead of 0 K, thermal effects can account for this discrepancy.

Table 1 and Fig. 1 also report experimental data of the enthalpy of formation for Cu_9Dy_2, Cu_2Dy and CuDy determined by Sommer et al. [20,30] by solution calorimetry and by Meschel et al. [21] by high-temperature direct calorimetry. The difference between ab initio and experimental values from Sommer et al. is significant. However, the experimental data from Meschel et al. are much closer to ab initio values. A similar discrepancy between experimental data sets from Sommer et al. and Meschel et al. also exists for the Cu–Er system, the former values being significantly less negative than the latter (Table 2). The ab initio results are also different from Sommer's data in other Cu–RE systems, as evidenced for some compounds in Table 2.

We have checked that our theoretical ab initio results do not depend significantly on the choice of the exchange–correlation functional. Recently, a systematic study of over 330 Al–RE compounds in 15 systems carried out using ab initio calculations and the VASP code has been performed [31]. These results confirm the validity of ab initio calculations for RE compounds. From this

Table 2
Comparison between experimental and ab initio values (calculated with both GGA and PBE potentials) for the enthalpy of formation for some Cu–RE compounds.

Compound	ΔH_f (kJ/mol of atoms)			Reference
	Exp.	GGA	PBE	
Cu_5Dy		−19.4	−18.8	This work
	−15.7			[21]
Cu_2Dy		−27.9	−26.1	This work
	−16.4			[20]
	−24.4			[21]
CuDy		−27.1	−25.5	This work
	−12.5			[20]
	−24.0			[21]
Cu_5Er		−20.1		This work
	−10.7			[20]
	−18.4			[21]
Cu_2Er		−29.1		This work
	−17.2			[20]
	−27.0			[21]
CuEr		−28.5		This work
	−14.6			[20]
CuGd		−26.4		This work
	−10.6			[20]

discussion, we conclude that experimental data from Sommer et al. must be considered with care.

Values obtained by Subramanian et al. using Miedema's semiempirical model [9] are also shown in Fig. 1 for comparison, and they are lower than both experimental and ab initio values. The values obtained according to the present thermodynamic assessment (full line in Fig. 1) are in good agreement with experimental data from Meschel et al. and with ab initio values. This further confirms that the stability of Cu–Dy compounds is higher than as measured by Sommer et al.

The phase diagram calculated according to the present thermodynamic assessment is reported in Fig. 2 together with experimental data from [8,10]. Calculated and experimental results for invariant reactions are summarized in Table 3. A general good agreement has been obtained except for the Dy-rich part of the diagram, which shows some disagreement with respect to the data by Franceschi. However, some uncertainties exist in this region of the system from the experimental point of view. In fact, it has already been mentioned that [14] reported a lower value for the eutectic temperature and a wide solubility range for Cu in bcc β-Dy with respect to Franceschi [8]. Experimental data on the enthalpy of mixing of the liquid phase are shown in Fig. 3 with the calculated line obtained from the thermodynamic assessment. An excellent agreement can be observed.

5. Conclusions

DFT ab initio calculations with the VASP code and plane waves have been performed on Cu–Dy intermetallic compounds. The calculated enthalpies of formation have been used to establish the relative stability of these phases at low temperature. The ab initio results have also been compared with experimental data. A satisfactory agreement has been obtained for lattice parameters and ab initio enthalpy of formation values were essential for discussing discrepancies between experimental calorimetry data.

By using ab initio results and experimental data, a complete set of optimized parameters has been obtained for the Cu–Dy system by applying the CALPHAD method. A good agreement has been obtained between experimental values and calculated results obtained from the assessed thermodynamic parameters. A higher discrepancy has been obtained in the Dy-rich side of the diagram, where experimental data are scarce and further experimental investigations are desirable. Due to insufficient experimental information available, the phase Cu_4Dy reported by [3] was not included in the present assessment, and it deserves further investigations.

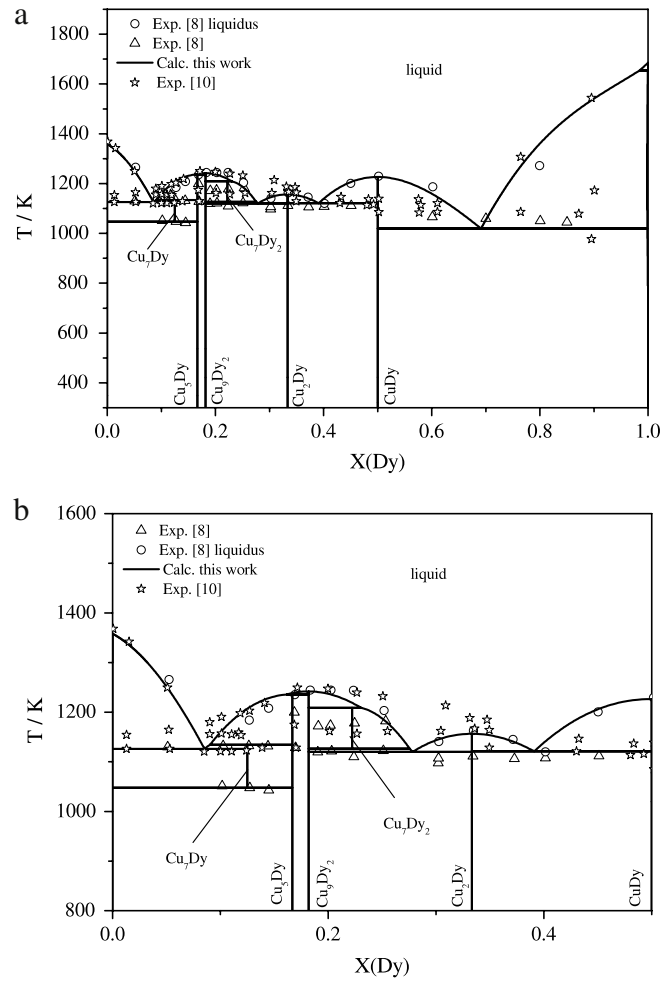

Fig. 2. (a) Calculated Cu–Dy phase diagram (full line) according to present thermodynamic description. Experimental points from Franceschi [8] and Cheng et al. [10] are also reported for comparison: circles, liquidus points from [8]; triangles, horizontal lines from [8]; stars, from [10]. (b) Enlargement of the Cu-rich part of the diagram.

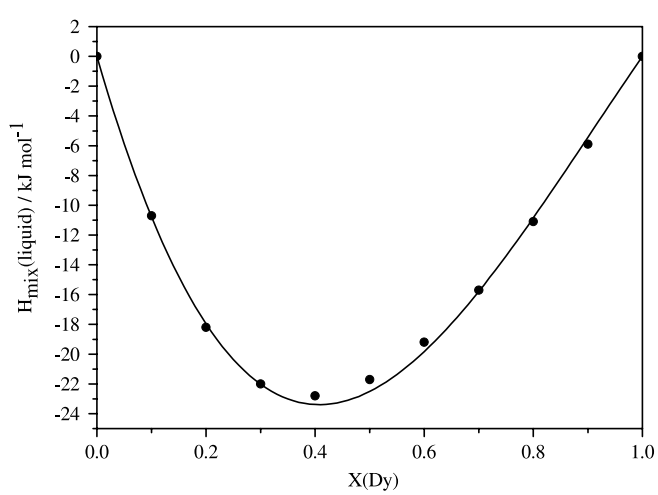

Fig. 3. Calculated (full line) and experimental (points) enthalpy of mixing of the liquid phase at 1690 K. Experimental data are from [18]. Reference states are Cu and Dy liquid phases at 1690 K.

Table 3
Calculated and experimental invariant reactions for the Cu–Dy system.

Reaction	Compositions of the respective phases (at.% Dy)			T (K)	Reference
liquid ↔ Cu fcc + Cu_7Dy	8.5	0	12.5	1126	This work
	10	0	12.5	1128	[8]
liquid + Cu_5Dy ↔ Cu_7Dy	9.0	16.7	12.5	1134	This work
	10.2	16.7	12.5	1165	[8]
	10.2	16.7	12.5	1133	[10]
Cu_7Dy ↔ Cu fcc + Cu_5Dy	12.5	0	16.7	1048	This work
	12.5	0	16.7	1048	[8]
	12.5	0	16.7	1121	[10]
liquid + Cu_9Dy_2 ↔ Cu_5Dy	16.2	18.2	16.7	1236	This work
	16.2	18.2	16.7	1238	[8]
liquid ↔ Cu_9Dy_2	18.2	18.2	-	1242	This work
	18.2	18.2	–	1243	[8]
liquid + Cu_9Dy_2 ↔ Cu_7Dy_2	23.1	18.2	22.2	1208	This work
	25.8	18.2	22.2	1178	[8]
Cu_7Dy_2 ↔ liquid + Cu_9Dy_2	22.2	27.6	18.2	1127	This work
	22.2	28.7	18.2	1128	[8]
liquid ↔ Cu_9Dy_2 + Cu_2Dy	27.8	18.2	33.3	1120	This work
	29.0	18.2	33.3	1118	[8]
liquid ↔ Cu_2Dy	33.3	33.3	–	1156	This work
	33.3	33.3	–	1163	[8]
liquid ↔ Cu_2Dy + CuDy	39.1	33.3	50	1121	This work
	39.5	33.3	50	1113	[8]
liquid ↔ CuDy	50	50	–	1227	This work
	50	50	–	1228	[8]
liquid ↔ CuDy + Dy bcc	69.1	50	1	1020	This work
	70	50	1	1063	[8]

Acknowledgment

The EU Network of Excellence "Complex Metallic Alloys" (CMA) project is acknowledged for supporting this work.

Appendix

Summary of optimized thermodynamic parameters for the Cu–Dy system. Parameters for pure elements are from [27].

*Liquid phase, model Redlich–Kister–Muggianu (*Cu, Dy*)*

$$^{0}L_{\mathrm{Cu,Dy}} \quad -90022.7625 - 2.09325615 * T \quad 298 < T < 3000$$

$$^{1}L_{\mathrm{Cu,Dy}} \quad -37038.1313 - 0.983676763 * T \quad 298 < T < 3000$$

Cu_2Dy *phase, model* $(Cu)_2(Dy)_1$

$$G^{\mathrm{Cu_2Dy}} \quad -25081.6021 - 1.40080692 * T + 0.6667 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.3333 * {}^{0}G^{\mathrm{hcp}}_{\mathrm{Dy}} \quad 298 < T < 3000$$

Cu_5Dy *phase, model* $(Cu)_5(Dy)_1$

$$G^{\mathrm{Cu_5Dy}} \quad -18889.5454 - 0.420383136 * T + 0.8333 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.1667 * {}^{0}G^{\mathrm{hcp}}_{\mathrm{Dy}} \quad 298 < T < 3000$$

Cu_7Dy *phase, model* $(Cu)_7(Dy)_1$

$$G^{\mathrm{Cu_7Dy}} \quad -13300 + 0.654306939 * T + 0.8333 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.1667 * {}^{0}G^{\mathrm{hcp}}_{\mathrm{Dy}} \quad 298 < T < 3000$$

Cu_7Dy_2 *phase, model* $(Cu)_7(Dy)_2$

$$G^{\mathrm{Cu_7Dy_2}} \quad -13020 - 8.235 * T + 0.7778 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.2222 * {}^{0}G^{hcp}_{\mathrm{Dy}} \quad 298 < T < 3000$$

Cu_9Dy_2 *phase, model* $(Cu)_9(Dy)_2$

$$G^{\mathrm{Cu_9Dy_2}} \quad -20472.8066 - 0.166118801 * T + 0.8182 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.1818 * {}^{0}G_{\mathrm{Dy}} \quad 298 < T < 3000$$

CuDy *phase, model* $(Cu)_1(Dy)_1$

$$G^{\mathrm{CuDy}} \quad -26977.5054 - 0.520388955 * T + 0.5 * {}^{0}G^{\mathrm{fcc}}_{\mathrm{Cu}} + 0.5 * {}^{0}G_{\mathrm{Dy}} \quad 298 < T < 3000$$

References

[1] Y. Itoh, H. Kadomatsu, J. Sakurai, H. Fujiwara, Phys. Status Solidi A 118 (1990) 513.
[2] F. Merlo, M. Pani, M.L. Fornasini, J. Alloys Compounds 221 (1995) 280.
[3] R. Černý, L. Guénée, R. Wessicken, J. Solid State Chem. 174 (2003) 125.
[4] R. Černý, M. François, K. Yvon, D. Jaccard, E. Walker, V. Petříček, I. Cisařová, H.-U. Nissen, R. Wessicken, J. Condens. Matter 8 (1996) 4485.
[5] S. Gottlieb-Schönmeyer, S. Brühne, F. Ritter, W. Assmus, S. Balanetskyy, M. Feuerbacher, T. Weber, W. Steurer, Intermetallics 17 (2009) 6.
[6] M. Feuerbacher, C. Thomas, S. Roitsch, Intermetallics 16 (2008) 943.
[7] P.E.A. Turchi, I.A. Abrikosov, B. Burton, S.G. Fries, G. Grimvall, L. Kaufman, P. Korzhavyi, V.R. Manga, M. Ohno, A. Pisch, A. Scott, W. Zhang, Calphad 31 (2007) 4.
[8] E. Franceschi, J. Less-Common Met. 87 (1982) 249.
[9] P.R. Subramanian, D.E. Laughlin, in: P.R. Subramanian, D.J. Chakrabarti, D.E. Laughlin (Eds.), Cu–Dy (Copper–Dysprosium) in Phase Diagrams of Binary Copper Alloys, ASM International, Materials Park, OH, 1994, p. 154.
[10] C.S. Cheng, G.-X. Xu, Acta Phys. Sinica 31 (6) (1982) 807.
[11] M.M. Carnasciali, G.A. Costa, E.A. Franceschi, J. Less-Common Met. 92 (1983) 143.
[12] K.H.J. Buschow, Philips J. Res. 25 (1970) 227.
[13] D.J. Chakrabarti, D.E. Laughlin, Bull. Alloy Phase Diagr. 2 (1981) 315.
[14] M. Copeland, H. Kato, Rare-earth-rich alloys, in: Physics and Materials Problems of Reactor Control Rods, International Atomic Energy Agency, Vienna, Austria, 1964, p. 295.
[15] K.H.J. Buschow, A.S. van der Goot, Acta Crystallogr. B 27 (6) (1971) 1085.
[16] K.H.J. Buschow, A.S. van der Goot, J. Birkhan, J. Less-Common Met. 19 (1969) 433.
[17] A.V. Tsvyashchenko, J. Less-Common Met. 99 (1984) L9.
[18] N.I. Usenko, M.I. Ivanov, V.M. Petiuh, V.T. Witusiewicz, J. Alloys Compounds 190 (1993) 149.
[19] V.S. Sudavtsova, G.I. Batalin, F.F. Kuznetsov, I.G. Bondarenko, Ukraïn. Khimicheskii Zh. 53 (1987) 498.
[20] F. Sommer, J. Schott, B. Predel, J. Less-Common Met. 125 (1986) 175.

[21] S.V. Meschel, O.J. Kleppa, J. Alloys Compounds 388 (2005) 91.
[22] G. Kresse, J. Hafner, Phys. Rev. B 47 (1993) 558.
[23] G. Kresse, J. Furthmuller, Comput. Mater. Sci. 6 (1996) 15.
[24] G. Kresse, J. Furthmuller, Phys. Rev. B 54 (1996) 11169.
[25] J.P. Perdew, Y. Wang, Phys. Rev. B 45 (1992) 13244.
[26] H.J. Monkhorst, J.D. Pack, Phys. Rev. B 13 (1976) 5188.
[27] N. Saunders, A.P. Miodownik, in: W. Cahn (Ed.), CALPHAD, Calculation of phase diagrams, in: Pergamon Materials Series, 1998.
[28] SGTE pure element database ver. 3.00 updated from: A. T. Dinsdale, Calphad 15 (1991) 317.
[29] O. Redlich, A. Kister, Indust. Eng. Chem. 40 (1948) 345.
[30] C. Colinet, A. Pasturel, in: K.A. Gschneidner Jr., L. Eyring, G.H. Lander, G.R. Choppin (Eds.), Handbook on the Physics and Chemistry of Rare Earths, vol. 19, Elsevier Science B.V., 1994, p. 479.
[31] M.C. Gao, A.D. Rollet, M. Widom, Phys. Rev. B 75 (2007) 174120.
[32] A.R. Storm, K.E. Benson, Acta Crystallogr. 16 (1963) 701.
[33] N.C. Baenziger, J.L. Moriarty Jr., Acta Crystallogr. 14 (1961) 948.
[34] C.C. Chao, H.L. Luo, P. Duwez, J. Appl. Phys. 35 (1964) 257.
[35] J. Pierre, R. Pauthenet, C.R. Hebd. Seances Acad. Sci. 260 (1965) 2739.
[36] A.E. Dwight, R.A. Conner Jr., J.W. Downey, in: Proc. 5th Rare Earth Res. Conf. Aug. 30–Sept. 1, 1965, Ames, LA, vol. 5, 1965, p. 35.
[37] M. Belakhovsky, J. Pierre, Solid State Commun. 9 (1971) 1409.